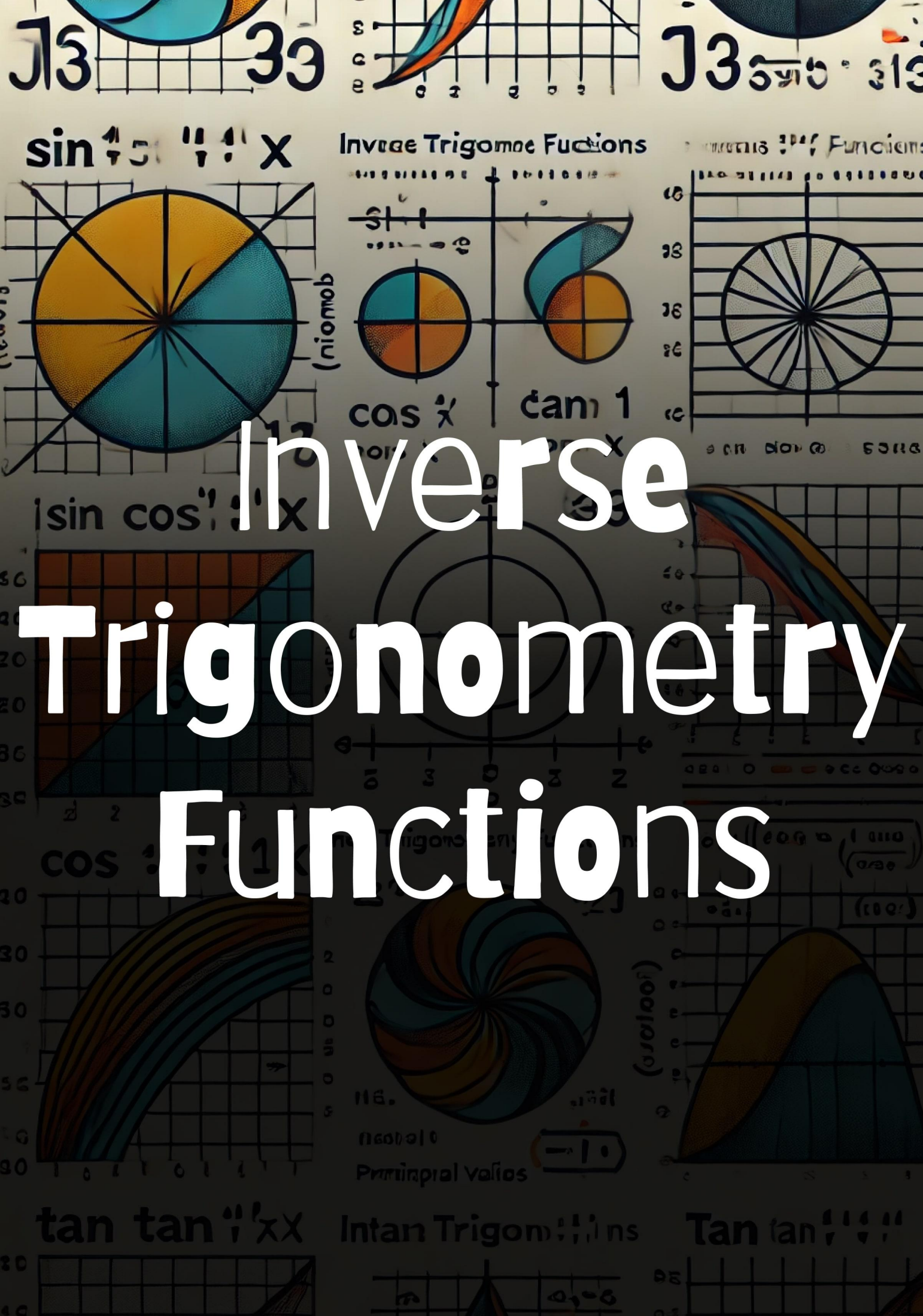

Inverse
Trigonometry
Functions

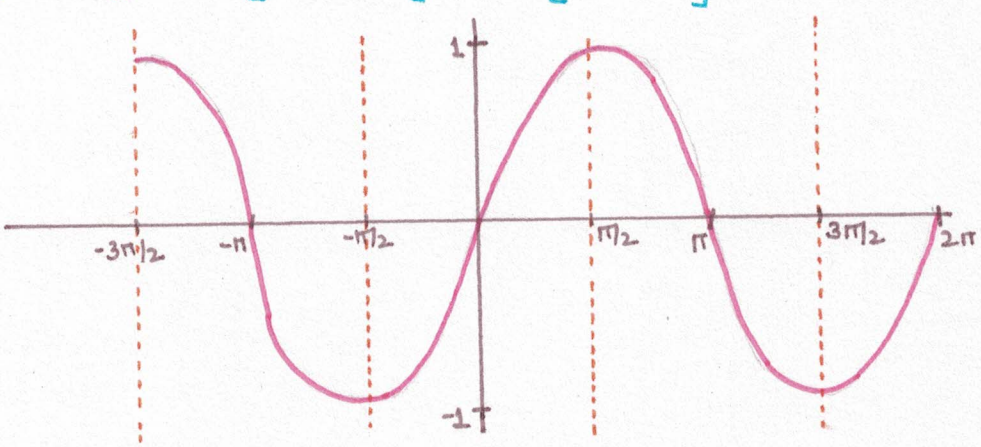
Inverse Trigonometry Functions

As we know, only bijective functions can have its inverse and every trigonometric function is a many-one functi-on. Hence, to obtain the inverse circular function, we have to restrict the domain and codomain of trigonometric function in such a way that they will become bijective.

Hence, we have to keep in mind that such kind of domain restrictions will have infinite ordered pairs, out of which we select numerically least ordered pair which provide total range.

For eg: $y = \sin x$, we can restrict its domain in

$\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right]$ or $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ or $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ or . . . . .



Here, numerically least ordered pair is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

It is also known as **Principle value or Principal range** of $\sin^{-1}x$.

Hence in ITF, we use 3 quardrants :-

I^st , IInd and IVth

...o points to remember o...

- For +ve value of x, all ITF will provide the output in I quardrant.
- If x is -ve, then $\sin^{-1}x$, $\tan^{-1}x$ and $\text{cosec}^{-1}x$ will provide output in IV quardrant whereas $\cos^{-1}x$, $\sec^{-1}x$, and $\cot^{-1}x$ will provide output in IInd quardrant
- $\sin^{-1}x$ is an angle. That's why $\sin^{-1}x$ cannot be equal to $\frac{1}{\sin x}$
- IIIrd quardrant cannot be used in ITF because, it will make ITF as many-one
- All ITF's are non-periodic bijective functions.

PROPERTIES OF ITF

PROPERTY-1

$\sin^{-1}x + \cos^{-1}x = \pi/2 \quad x \in [-1,1]$

$\tan^{-1}x + \cot^{-1}x = \pi/2 \quad x \in R$

$\sec^{-1}x + \text{cosec}^{-1}x = \pi/2 \quad x \in (-\infty,-1] \cup [1,\infty)$

Proof: $\sin^{-1}x = \theta \qquad \theta = [-\pi/2, \pi/2]$

$x = \sin\theta$

$x = \cos\left(\frac{\pi}{2} - \theta\right) \qquad \pi/2 - \theta \in [0, \pi]$

$\cos^{-1}x = \frac{\pi}{2} - \theta$

$\cos^{-1}x + \theta = \frac{\pi}{2}$

$$\sin^{-1}x + \cos^{-1}x = \pi/2$$

Ques: Find the range of $(\sin^{-1}x)^2 + (\cos^{-1}x)^2$

Sol:

$$y = (\sin^{-1}x)^2 + (\cos^{-1}x)^2$$

$$y = (\sin^{-1}x + \cos^{-1}x)^2 - 2\sin^{-1}x \cdot \cos^{-1}x$$

$$y = \left(\frac{\pi}{2}\right)^2 - 2\sin^{-1}x\left(\frac{\pi}{2} - \sin^{-1}x\right)$$

$$y = \frac{\pi^2}{4} - \pi\sin^{-1}x + 2(\sin^{-1}x)^2$$

$$y = 2\left[(\sin^{-1}x)^2 - \pi/2\,(\sin^{-1}x) + \pi^2/8\right]$$

$$y = 2\left[\sin^{-1}x - \pi/4\right]^2 + \pi^2/8$$

$$\text{Range}(y) = \left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$$

Ques: Find the range of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$

Sol:

$$y = (\sin^{-1}x)^3 + (\cos^{-1}x)^3$$

$$y = (\sin^{-1}x + \cos^{-1}x)^3 - 3\sin^{-1}x\cos^{-1}x(\sin^{-1}x + \cos^{-1}x)$$

$$y = \frac{\pi^3}{8} - 3\sin^{-1}x\left(\frac{\pi}{2} - \sin^{-1}x\right)\left(\frac{\pi}{2}\right)$$

$$y = \frac{\pi^3}{8} - \frac{3\pi^2}{4}\sin^{-1}x + \frac{3\pi}{2}(\sin^{-1}x)^2$$

$$y = \frac{3\pi}{2}\left[(\sin^{-1}x)^2 - \frac{\pi}{2}\sin^{-1}x\right] + \frac{\pi^3}{8} - \frac{3\pi^3}{32}$$

$$y = \frac{3\pi}{2}\left[\sin^{-1}x - \frac{\pi}{4}\right]^2 + \frac{\pi^3}{32}$$

$$\text{Range}(y) = \left[\frac{\pi^3}{32}, \frac{7\pi^3}{8}\right]$$

Ques: Find the range of $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$

Sol:

$\sin^{-1}x + \cos^{-1}x = \pi/2 \qquad x \in [-1, 1]$

$\pi/2 + \tan^{-1}x$

$\pi/2 - \pi/4$ to $\pi/2 + \pi/4$

$$\text{Range}(y) = \pi/4 \text{ to } 3\pi/4$$

Ques: $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$, then find:

$$x^{2016} + y^{2016} + z^{2016} - \frac{3}{x^{2016}+y^{2016}+z^{2016}} = ?$$

Sol: $\sin^{-1}x = \sin^{-1}y = \sin^{-1}z = \pi/2$

$x = y = z = 1$

putting values in expression.

$$1 + 1 + 1 - \frac{3}{3}$$

Expression = 2

PROPERTY-2

$\sin^{-1}(-x) = -\sin^{-1}x \quad x \in [-1,1]$

$\cos^{-1}(-x) = \pi - \cos^{-1}x \quad x \in [-1,1]$

$\tan^{-1}(-x) = -\tan^{-1}x \quad x \in R$

$\cot^{-1}(-x) = \pi - \cot^{-1}x \quad x \in R$

$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x \quad x \in R-(-1,1)$

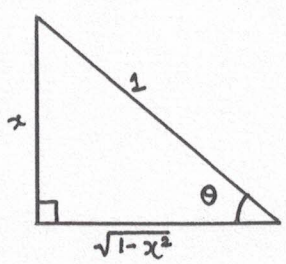
$\sec^{-1}(-x) = \pi - \sec^{-1}x \quad x \in R-(-1,1)$

PROPERTY-3

Conversion Formula

$\theta = \sin^{-1}x \quad x \geq 0$

$\sin\theta = x \quad \theta \in \left[0, \frac{\pi}{2}\right]$



$\cos\theta = \sqrt{1-x^2} \quad \theta \in \left[0, \frac{\pi}{2}\right]$

$\theta = \cos^{-1}\sqrt{1-x^2}$

$\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$

$\tan\theta = \frac{x}{\sqrt{1-x^2}} \quad \theta \in \left[0, \frac{\pi}{2}\right]$

$\sin^{-1}x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

Similarly,

$\sin^{-1}x = \cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

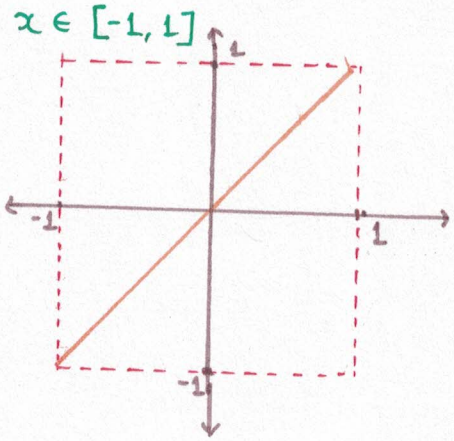
$\sin^{-1}x = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$

$$\sin^{-1}x = \operatorname{cosec}^{-1}\left(\frac{1}{x}\right)$$

TRY YOURSELF Convert $\cos^{-1}x$ into remaining 5 as well as $\tan^{-1}x$

PROPERTY-4

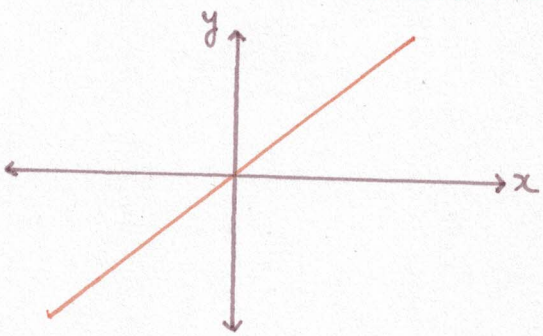
⊙ $\sin(\sin^{-1}x) = x$ $\quad x \in [-1, 1]$



⊙ $\cos(\cos^{-1}x) = x$ $\quad x \in [-1, 1]$

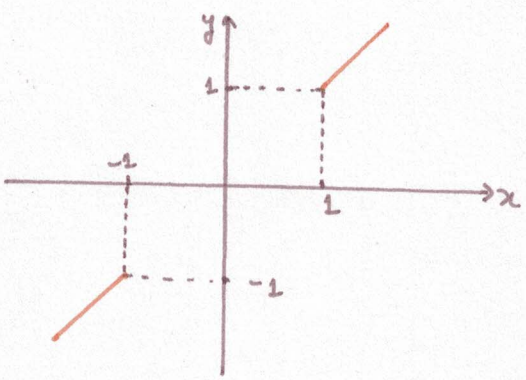
⊙ $\tan(\tan^{-1}x) = x$ $\quad x \in R$

⊙ $\cot(\cot^{-1}x) = x$ $\quad x \in R$



⊙ $\operatorname{cosec}(\operatorname{cosec}^{-1}x) = x$ $\quad x \in R - (-1, 1)$

⊙ $\sec(\sec^{-1}x) = x$ $\quad x \in R - (-1, 1)$



Ques: Find the value of $\sin\left(\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right)$

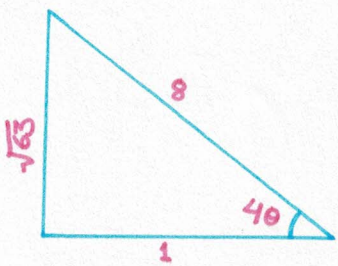
Sol: $\sin\left(\sin^{-1}\frac{3}{5}\right)\cos\left(\cos^{-1}\frac{5}{13}\right) + \cos\left(\sin^{-1}\frac{3}{5}\right)\sin\left(\cos^{-1}\frac{5}{13}\right)$

$$= \frac{3}{5}\times\frac{5}{13} + \frac{4}{5}\times\frac{12}{13} = \frac{63}{65}$$

Ques: Find the value of $\sin\left(\frac{1}{4}\left(\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)\right)\right)$

Sol: Let $\sin^{-1}\frac{\sqrt{63}}{8} = 4\theta$ $\quad 4\theta \in \left(0, \frac{\pi}{2}\right)$

$\sin 4\theta = \frac{\sqrt{63}}{8}$ $\quad \cos 4\theta = \frac{1}{8}$



$2\cos^2 2\theta - 1 = \frac{1}{8}$

$2\cos^2 2\theta = \frac{9}{8}$

$\cos^2 2\theta = \frac{9}{16}$

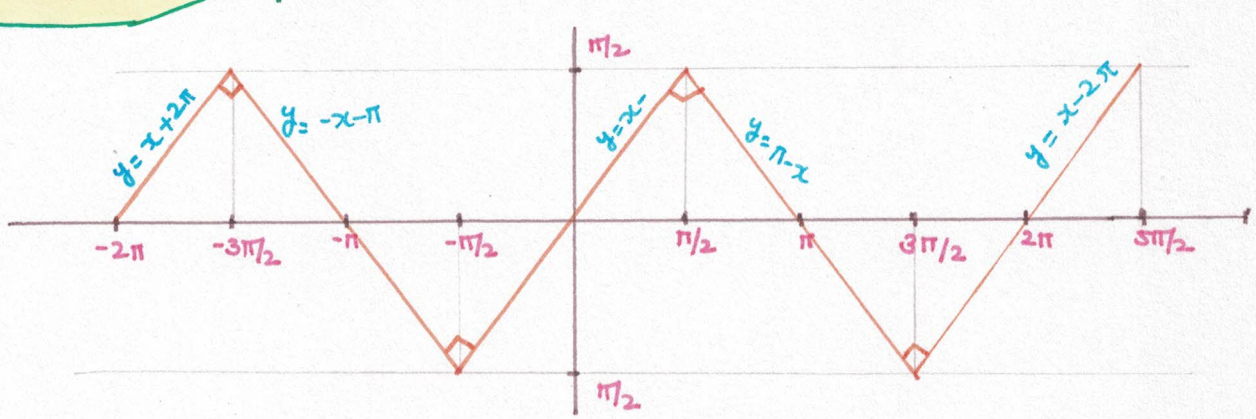
$\cos 2\theta = \frac{3}{4}$ ✓ $\quad \cos 2\theta = -\frac{3}{4}$ ✗ $\quad 2\theta \in \left(0, \frac{\pi}{8}\right)$

$1 - 2\sin^2\theta = \frac{3}{4}$

$\sin^2\theta = \frac{1}{8}$

$\sin\theta = \frac{1}{2\sqrt{2}}$ ✓ $\quad \sin\theta = \frac{-1}{2\sqrt{2}}$ ✗ $\quad \theta \in \left(0, \frac{\pi}{8}\right)$

PROPERTY-5 i) $\sin^{-1}(\sin x)$ $\quad x \in R$



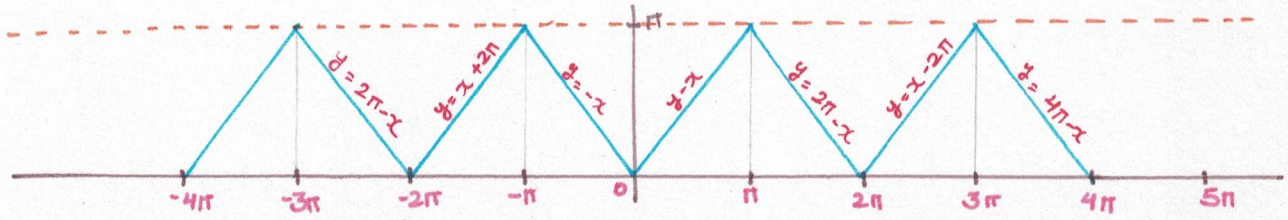
It is a periodic function with the period 2π

Ques: $\sin^{-1}(\sin 4) + \sin^{-1}(\sin 5) + \sin^{-1}(\sin 8)$

Sol: $\pi - 4$ $\quad 5 - 2\pi$ $\quad 3\pi - 8$

$= 2\pi - 7$

ii) $\cos^{-1}(\cos x)$ $\quad x \in R$



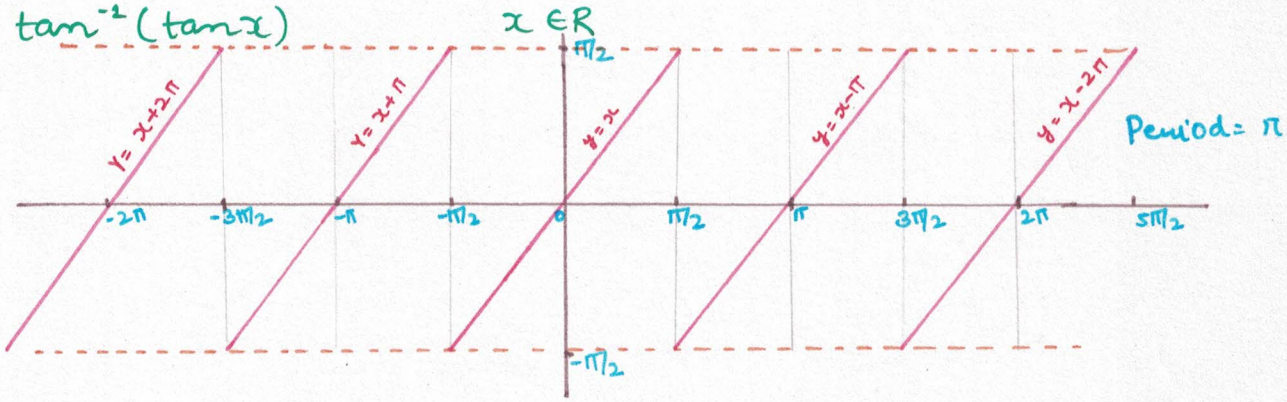
It is a periodic function with the period 2π.

Ques: Find the value of $\cos^{-1}(\cos 100)$

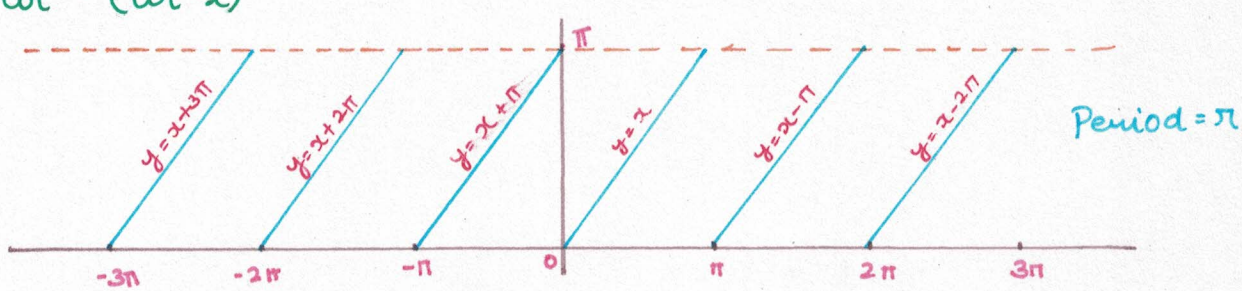
Sol: $x = 100$

value $= 32\pi - 100$

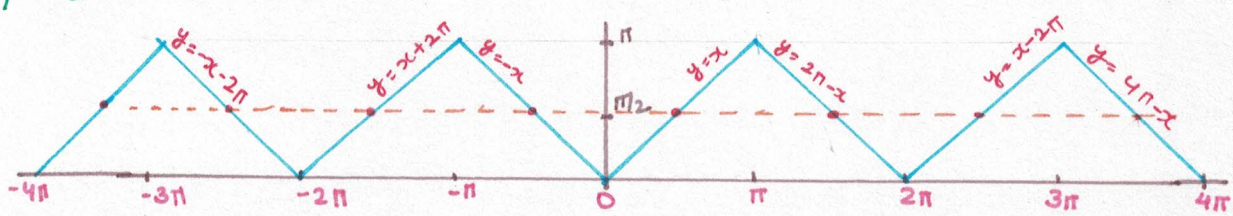
iii) $\tan^{-1}(\tan x)$ $\quad x \in R$



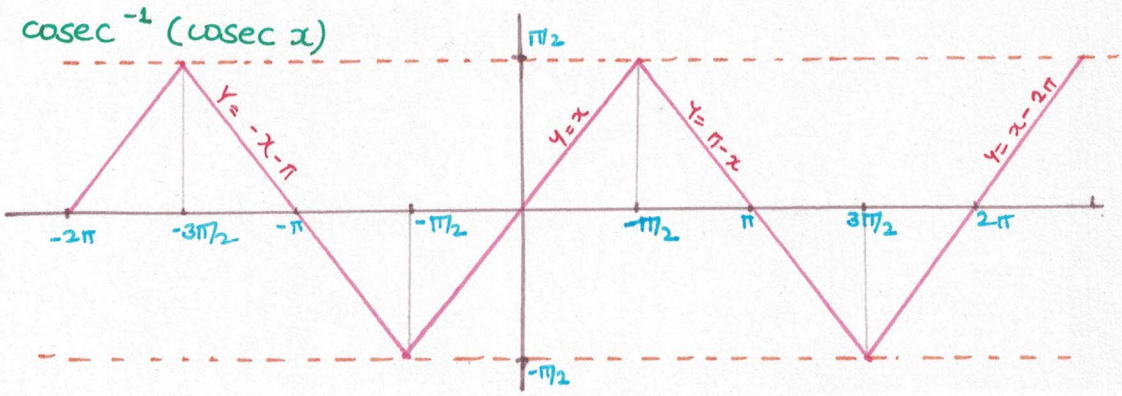
iv) $\cot^{-1}(\cot x)$



v) $\sec^{-1}(\sec x)$



vi) $\operatorname{cosec}^{-1}(\operatorname{cosec} x)$



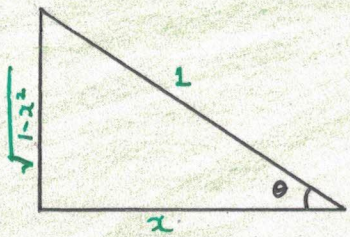
$\cos^{-1}x$

$\cos^{-1}x = \sin^{-1}\sqrt{1-x^2}$

$\cos^{-1}x = \operatorname{cosec}^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$

$\cos^{-1}x = \sec^{-1}\left(\frac{1}{x}\right)$

$\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$, $\cos^{-1}x = \cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$



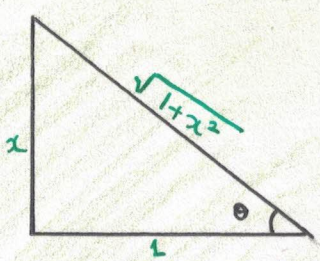
$\tan^{-1}x$

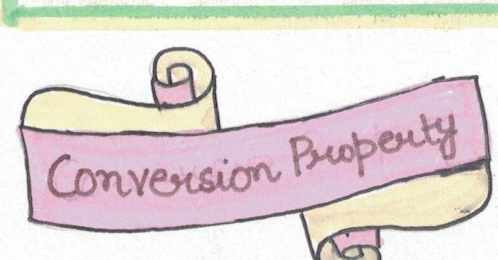
$\tan^{-1}x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$

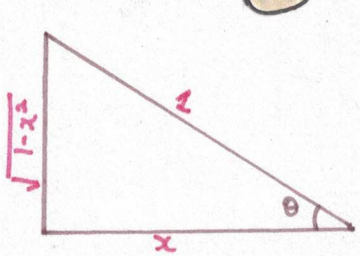
$\tan^{-1}x = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

$\tan^{-1}x = \cot^{-1}\left(\frac{1}{x}\right)$

$\tan^{-1}x = \sec^{-1}\left(\sqrt{1+x^2}\right)$, $\tan^{-1}x = \operatorname{cosec}^{-1}\left(\frac{\sqrt{1+x^2}}{x}\right)$







$\cos^{-1}x = \theta \qquad x < 0$

$\Rightarrow \theta = \cos^{-1}x \qquad x \in (\pi/2, \pi)$

$\Rightarrow x = \cos\theta$

$\Rightarrow \sin\theta = \sqrt{1-x^2} \qquad x \in (\pi/2, \pi)$

$\Rightarrow \sin^{-1}(\sin\theta) = \sin^{-1}(\sqrt{1-x^2})$

$\Rightarrow \pi - \theta = \sin^{-1}(\sqrt{1-x^2})$

$\theta = \pi - \sin^{-1}(\sqrt{1-x^2})$

$\tan\theta = \frac{\sqrt{1-x^2}}{x} \qquad x \in (\pi/2, \pi)$

$\Rightarrow \tan^{-1}(\tan\theta) = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$\Rightarrow \theta - \pi = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$\Rightarrow \theta = \pi + \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$$\theta = \pi - \operatorname{cosec}^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$$

$$\theta = \sec^{-1} x$$

$$\theta = \cot^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$$

PROPERTY 6

$$\begin{cases} \sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1}(x) \\ \operatorname{cosec}^{-1}\left(\frac{1}{x}\right) = \sin^{-1}(x) \end{cases}$$

$x \in (-\infty, -1] \cup [1, \infty)$

$x \in [-1, 1] - \{0\}$

$$\begin{cases} \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1}(x) \\ \sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1}(x) \end{cases}$$

$x \in (-\infty, -1] \cup [1, \infty)$

$x \in [-1, 1] - \{0\}$

$$\cot^{-1}\left(\frac{1}{x}\right) = \begin{cases} \tan^{-1} x & x > 0 \\ x + \tan^{-1} x & x < 0 \end{cases}$$

$\theta \in (0, \pi)$

$$\cot^{-1}\left(\frac{1}{x}\right) = \theta$$

$$\frac{1}{x} = \cot\theta$$

$$\tan\theta = x$$

$$\tan^{-1}\tan\theta = \tan^{-1} x$$

CASE-I

$0 < \theta < \pi/2$

$0 < \cot^{-1}\left(\frac{1}{x}\right) < \pi/2$

$\cot 0 > \frac{1}{x} > \cot \pi/2$

$\infty > \frac{1}{x} > 0$

$0 < x < \infty$

$x > 0$

CASE-II

$\pi/2 < \theta < \pi$

$\pi/2 < \cot^{-1}\left(\frac{1}{x}\right) < \pi$

$\cot \pi/2 > \frac{1}{x} > \cot \pi$

$0 > \frac{1}{x} > -\infty$

$0 > x > -\infty$

$x < 0$

Ques: $\sin\left(\sin^{-1}\frac{1}{3} + \sec^{-1}\frac{3}{2\sqrt{2}}\right)$

Sol: $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{1}{3}\right)\left(\frac{2\sqrt{3}}{3}\right) = \frac{4\sqrt{2}}{9}$

FORMULAS

1. $\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & x>0 \quad y>0 \\ \pi/2 & x>0 \quad y>0 \\ x + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & x>0 \quad y>0 \end{cases}$

PROOF: Let $\tan^{-1}x = A$ $\quad \tan^{-1}y = B$ $\quad x>0, y>0$

$x = \tan A \qquad y = \tan B$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan^{-1}\tan(A+B) = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \qquad A+B \in (0, \pi)$$

CASE-1 $0 < A+B < \frac{\pi}{2}$

$$A+B = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$1-xy > 0 \quad \Rightarrow \quad xy < 1$

CASE-2 $\frac{\pi}{2} < A+B < \pi$

$$A+B-\pi = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$1-xy < 0 \quad \Rightarrow \quad xy > 1$

2. $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right) \qquad x \geq 0, \; y>0$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$A \in (0, \frac{\pi}{2})$

$B \in (0, \frac{\pi}{2})$

$\tan \tan^{-1}(A-B) = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$ $\quad A-B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$A-B = \tan^{-1}\frac{x-y}{1+xy}$

$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$

Ques: $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) = ?$

Sol: $xy > 1$ $\quad \pi + \tan^{-1}\left(\frac{3}{1-2}\right) + \tan^{-1}(3)$

$= \pi + \tan^{-1}(-3) + \tan^{-1}(3)$

$= \pi - \tan^{-1}(3) + \tan^{-1}(3)$

$= \pi$

Ques: $\tan^{-1}(1) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$

Sol: $xy < 1$ $\quad \tan^{-1}\left(\frac{3/2}{1-1/2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$

$xy = 1$ $\quad \tan^{-1}(3) + \tan^{-1}\left(\frac{1}{3}\right)$

$= \pi/2$

Ques: $\tan^{-1}(2) + \tan^{-1}(3) - \tan^{-1}(1/2) + \tan^{-1}(-1/3)$

Sol: $\pi + \tan^{-1}\left(\frac{5}{-5}\right) - \tan^{-1}\left(\frac{1}{2}\right)$

$= \pi - \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{3}\right)$

$xy < 1$ $\quad \pi - \tan^{-1}\left(\frac{3/2}{1-1/2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$

$= \pi - \tan^{-1}3 + \tan^{-1}\left(\frac{1}{3}\right)$

$= \pi - \pi/2$

$= \pi/2$

Ques: Evaluate the formula of $2\tan^{-1}x$?

Sol: $2\tan^{-1}x = \tan^{-1}x + \tan^{-1}x$

$\Rightarrow \tan^{-1}(\tan 2x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$$2\tan^{-1}x = \begin{cases} -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & -\infty < x < -1 \\ \tan^{-1}\left(\frac{2x}{1-x^2}\right) & -1 < x < 1 \\ \pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & 1 < x < \infty \\ -\pi/2 & x = -1 \\ \pi/2 & x = 1 \end{cases}$$

$\Rightarrow \tan^{-1}x = A \qquad A \in \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$

$\Rightarrow x = \tan A$

$\Rightarrow \tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

$\Rightarrow \tan^{-1}\tan 2A = \tan^{-1}\left(\dfrac{2\tan A}{1-\tan^2 A}\right) \qquad 2A \in (-\pi, \pi)$

CASE-1

$-\pi < A < \pi/2$

$\tan^{-1}x = -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$-\infty < x < -1$

CASE-2

$-\pi/2 < A < \pi/2$

$\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$-1 < x < 1$

CASE-3

$\pi/2 < A < \pi$

$\tan^{-1}x = \pi + \tan^{-1}\left(\frac{2x}{1-x}\right)$

$1 < x < \infty$

$$3\tan^{-1}x = \begin{cases} -\pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) & -\infty < x < -1/\sqrt{3} \\ -\pi/2 & x = -1/\sqrt{3} \\ \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) & -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ \pi/2 & x = 1/\sqrt{3} \\ \pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) & 1/\sqrt{3} < x < \infty \end{cases}$$

$\sin^{-1}x + \sin^{-1}y \quad x \geq 0 \quad y \geq 0$

$\sin^{-1}x = A \qquad \sin^{-1}y = B \qquad A, B \in \left[0, \frac{\pi}{2}\right]$

$x = \sin A \qquad y = \sin B$

$\sin(A+B) = \sin A\cos B + \cos A \sin B$

$\sin^{-1}\sin(A+B) = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) \qquad 0 \leq A+B \leq \pi$

Case-1: $0 \le A+B \le \pi/2$

$A+B = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

$\sin^{-1}x + \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

$A+B \le \frac{\pi}{2} \Rightarrow A \le \frac{\pi}{2} - B \Rightarrow \sin A \le \sin\left(\frac{\pi}{2} - B\right)$

$\Rightarrow \sin A \le \cos B$

$\Rightarrow x \le \sqrt{1-y^2}$

$$\boxed{x^2+y^2 \le 1}$$

Case-2: $\frac{\pi}{2} < A+B \le \pi$

$\pi - (A+B) = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

$A+B = \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

$A > \frac{\pi}{2} - B$

$\sin A > \cos B$

$$\boxed{x^2+y^2 > 1}$$

$$\sin^{-1}x + \sin^{-1}y = \begin{cases} \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & x\ge 0,\ y\ge 0,\ x^2+y^2 \le 1 \\ \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & x\ge 0,\ y\ge 0,\ x^2+y^2 > 1 \end{cases}$$

$$\boxed{\sin^{-1}x - \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2}) \quad x\ge 0,\ y\ge 0}$$

$\cos^{-1}x + \cos^{-1}y \quad x\ge 0,\ y\ge 0$

$\cos^{-1}x = A \qquad \cos^{-1}y = B$

$A \in \left(0, \frac{\pi}{2}\right), \quad B \in \left(0, \frac{\pi}{2}\right)$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos(A+B) = (xy - \sqrt{1-x^2}\sqrt{1-y^2})$

$\cos^{-1}\cos(A+B) = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2}) \qquad A+B \in (0, \pi)$

$$\boxed{\cos^{-1}x + \cos^{-1}y = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2} \quad x\ge 0,\ y\ge 0}$$

$\cos^{-1}x - \cos^{-1}y \quad x\ge 0,\ y\ge 0$

$\cos^{-1}x = A \qquad \cos^{-1}y = B$

$x = \cos A$ $\quad y = \cos B$

$\cos(A-B) = \cos A \cos B - \sin A \sin B$

$\cos^{-1}\cos(A-B) = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$ $\quad (A-B) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Case 1 $\quad A-B \in \left[-\frac{\pi}{2}, 0\right]$

$A-B = -\cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$

$A-B \le 0$

$A \le B$

$\cos^{-1}x \le \cos^{-1}y$

$x \ge y$

Case 2 $\quad A-B \in \left[0, \frac{\pi}{2}\right]$

$A-B = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$

$A-B \ge 0$

$A \ge B$

$\cos^{-1}x \ge \cos^{-1}y$

$x \le y$

Ques: $\cos\left[\frac{1}{2}\cos^{-1}\frac{1}{8}\right]$

Sol: $\cos^{-1}\frac{1}{8} = \theta$ $\quad \cos\left(\frac{\theta}{2}\right)$ $\quad 2\theta \in \left(0, \frac{\pi}{2}\right)$

$\frac{1+\cos 2\theta}{2} = \cos^2\theta$ $\quad \frac{1+\cos\theta}{2} = \cos^2\frac{\theta}{2}$

$\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$

$\cos\frac{\theta}{2} = \frac{3}{4}$

Ques: $[\sin^{-1}x] + [\cos^{-1}x] = 0$ $\quad$ Find value of x, when x is non-ve.

Sol: $[-2 \text{ to } 1] + [0 \text{ to } 3]$

$0 \le \sin^{-1}x < 1$ $\quad 0 \le \cos^{-1}x < 1$

$0 \le x < \sin 1$ $\quad \cos 1 \le x \le 1$

$x \ne -2$
$x \ne -1$
$x = 0$

Intersection : $(\cos 1, \sin 1)$

$$\cos^{-1}x - \cos^{-1}y = \begin{cases} -\cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2} & x \ge y \\ \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2} & y \ge x \end{cases}$$

Ques: Convert $2\tan^{-1}x$ in $\sin^{-1}$

Sol: $\tan^{-1}x = \theta \qquad x \in (-\pi/2, \pi/2)$

$x = \tan\theta$

$\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta} \qquad 2\theta \in (-\pi, \pi)$

$\sin^{-1}\sin 2\theta = \sin^{-1}\dfrac{2x}{1+x^2}$

Case 1

$-\pi < 2\theta < -\pi/2$

$-\pi/2 < \theta < -\pi/4$

$-\infty < x < -1$

$2\theta = -\pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

Case 2

$-\pi/2 \le 2\theta < \pi/2$

$-\pi/4 \le \theta < \pi/4$

$-1 \le x < 1$

$2\theta = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

Case 3

$\pi/2 \le 2\theta < \pi$

$\pi/4 < \theta < \pi/2$

$1 \le x < \infty$

$2\theta = \pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

Ques: Find the value of $\cos^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{-1}{7}\right) + \cos^{-1}\left(\frac{13}{14}\right)$

Sol: $\dfrac{\pi}{3} + \pi - \cos^{-1}\left(\dfrac{1}{7}\right) + \cos^{-1}\left(\dfrac{13}{14}\right)$

$\Rightarrow \dfrac{\pi}{3} + \pi - \cos^{-1}\left[\dfrac{13}{14}\times\dfrac{1}{7} + \dfrac{3\sqrt{3}}{14}\times\dfrac{4\sqrt{3}}{7}\right]$

$\Rightarrow \dfrac{\pi}{3} + \pi - \cos^{-1}\left[\dfrac{13+36}{14\times 7}\right]$

$\Rightarrow \dfrac{\pi}{3} + \pi - \cos^{-1}\left(\dfrac{1}{2}\right)$

$\Rightarrow \dfrac{\pi}{3} + \pi - \dfrac{\pi}{3}$

$\Rightarrow \pi$

Inverse Trignometric Equations

Ques: Find the value of x so that $2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

Sol: $x = \cos\theta \qquad \theta \in [0, \pi]$

$\Rightarrow \sin 2\theta = 2x\sqrt{1-x^2} \qquad \because \sin 2\theta = 2\sin\theta\cos\theta$

$2\theta \in [0, 2\pi]$

$\Rightarrow \sin^{-1}\sin 2\theta = \sin^{-1}(2x\sqrt{1-x^2})$

$0 \le 2\theta \le \dfrac{\pi}{2}$

$0 \le \theta \le \dfrac{\pi}{4}$

$2\theta = \sin^{-1}(2x\sqrt{1-x^2})$

$2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

$\Rightarrow \quad 0 \leq \cos^{-1}x \leq \pi/4$

$\Rightarrow \quad \cos^{-1}0 \geq x \geq \cos^{-1}\pi/4$

$\Rightarrow \quad 1 \geq x \geq \frac{1}{\sqrt{2}}$

$$\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & xy<1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & xy>1 \end{cases}$$

$$\sin^{-1}x + \sin^{-1}y = \begin{cases} \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) & x^2+y^2 \leq 1 \\ \pi - \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) & x^2+y^2>1 \end{cases}$$

$$\cos^{-1}x - \cos^{-1}y = \begin{cases} \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) & y \geq x \\ -\cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) & x > y \end{cases}$$

Ques: $\tan^{-1}\left(\frac{x+1}{x-1}\right) + \tan^{-1}\left(\frac{x-1}{x}\right) = \tan^{-1}(-7)$

Sol: $\tan^{-1}\left(\frac{\frac{x+1}{x-1} + \frac{x-1}{x}}{1-\left(\frac{x+1}{x-1}\right)\left(\frac{x-1}{x}\right)}\right)$

$\Rightarrow \quad \tan^{-1}\left(\frac{x^2+x+x^2+1-2x}{x^2-x-(x^2-1)}\right)$

$\Rightarrow \quad \tan^{-1}\left(\frac{2x^2-x+1}{1-x}\right) = \tan^{-1}(-7)$

$\Rightarrow \quad \frac{2x^2-x+1}{x-1} = 7$

$\Rightarrow \quad 2x^2-x+1 = 7x-7$

$\Rightarrow \quad 2x^2-8x+8 = 0$

$\Rightarrow \quad (x-2)^2 = 0 \qquad x=2$

But x should be less than zero, hence no solution.

IMPORTANT

Summation of Series Containing ITF

In order to obtain the summation of series containing ITF, use following steps:

1. Find the expression of nth term of the series i.e. T_n
2. If the series containing ITF other then $\tan^{-1}$, then convert the general term in the form of $\tan^{-1}$
3. Split the general term in 2 parts such that both parts will form the formula of $\tan^{-1}x - \tan^{-1}y$ of consecutive terms.

Now use summation or Σ function so that all the terms will cancel except first and last or second and second last, which is our required selection.

Ques: $\cot^{-1}3 + \cot^{-1}7 + \cot^{-1}13 + \cot^{-1}21 + \ldots\ldots$ n times = ?

sol: $T_n = \cot^{-1}(n^2+n+1)$

$S_n = 3+7+13+21+\ldots\ldots\ldots

'0' Difference AP	$T_n = an+b$
'Ist' Difference AP	$T_n = an^2+bn+c$
'2nd' Difference AP	$T_n = an^3+bn^2+cn+d$

Ques: $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{1}{18}\right) + \dots\dots\dots \infty$

Sol: $T_n = \tan^{-1}\left(\frac{1}{2n^2}\right) = \tan^{-1}\left(\frac{2}{4n^2}\right)$

$\Rightarrow T_n = \tan^{-1}\left(\frac{2}{1+4n^2-1}\right) = \tan^{-1}\left(\frac{2}{1+(2n-1)(2n+1)}\right)$

$\Rightarrow T_n = \tan^{-1}\left(\frac{(2n+1)-(2n-1)}{1+(2n-1)(2n+1)}\right)$

$\Rightarrow T_n = \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$

$\Rightarrow \sum_{n=1}^{n} T_n = \sum_{n=1}^{n} \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$

$\Rightarrow = \tan^{-1}(3) - \tan^{-1}(1)$
$+ \tan^{-1}(5) - \tan^{-1}(3)$
$+ \tan^{-1}(7) - \tan^{-1}(5)$
$\vdots \qquad \vdots$
$\tan^{-1}(2n+1) \qquad \tan^{-1}(2n-1)$

$\Rightarrow \sum_{n=1}^{n\to\infty} T_n = \tan^{-1}(2n+1) - \pi/4$

$\Rightarrow = \tan^{-1}(\infty) - \pi/4$

$\Rightarrow = \pi/2 - \pi/4$

$\Rightarrow = \pi/4$



Ques: $\tan^{-1}\left(\frac{4}{1+3.4}\right) + \tan^{-1}\left(\frac{6}{1+8.9}\right) + \tan^{-1}\left(\frac{8}{1+15.16}\right) + \dots\dots \infty$

Sol: $\tan^{-1}\left(\frac{6-2}{1+3.4}\right) + \tan^{-1}\left(\frac{12-6}{1+8.9}\right) + \tan^{-1}\left(\frac{20-12}{1+15.16}\right) + \dots\dots\infty$

$\Rightarrow \tan^{-1}\left(\frac{6-2}{1+6.2}\right) + \tan^{-1}\left(\frac{12-6}{1+12.6}\right) + \tan^{-1}\left(\frac{20-12}{1+20.12}\right) + \dots\dots\infty$

$\Rightarrow$ $\tan^{-1}6 - \tan^{-1}2$

$+ \tan^{-1}12 - \tan^{-1}6$

$+ \tan^{-1}20 - \tan^{-1}12$

$\vdots \qquad \vdots$

$\tan^{-1}(n^2+3n+2) - \tan^{-1}(n^2+n)$

$\Rightarrow \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$

$n \to \infty$

$\Rightarrow \tan^{-1}(\infty) - \tan^{-1}\left(\frac{1}{2}\right)$

$\Rightarrow \boxed{Z = \frac{\pi}{2} - \cot^{-1}(2)}$

Ques: $\tan^{-1}\left(\frac{4}{1+3\cdot4}\right) + \tan^{-1}\left(\frac{6}{1+8\cdot9}\right) + \tan^{-1}\left(\frac{8}{1+15\cdot16}\right) + \dots \infty$

Sol: $\tan^{-1}\left(\frac{6-2}{1+3\cdot4}\right) + \tan^{-1}\left(\frac{12-6}{1+8\cdot9}\right) + \tan^{-1}\left(\frac{20-12}{1+15\cdot16}\right) + \dots \infty$

$\Rightarrow \tan^{-1}\left(\frac{6-2}{1+6\cdot2}\right) + \tan^{-1}\left(\frac{12-6}{1+12\cdot6}\right) + \tan^{-1}\left(\frac{20-12}{1+20\cdot12}\right) + \dots \infty$

$\Rightarrow \tan^{-1}6 - \tan^{-1}2$

$\tan^{-1}12 - \tan^{-1}6$

$\tan^{-1}20 - \tan^{-1}12$

$\vdots \qquad \vdots$

$\tan^{-1}(n^2+3n+2) - \tan^{-1}(n^2+n)$

$\Rightarrow \tan^{-1}(n^2+3n+2) - \tan^{-1}2$

$n \to \infty$

$\Rightarrow \boxed{\text{Sum} \to \frac{\pi}{2} - \tan^{-1}(2)}$